1. MC.11.01.ALGO (Algorithmic)

The practice of delegating authority to division-level managers by top management is

a. reorganization.

b. management span of control.

✓ c. decentralization.

d. centralization.

e. only done in manufacturing businesses.

Feedback

Check My Work

Centralized decision making is decisions being made at the very top level, and lower-level managers being responsible for implementing these decisions. Decentralization is the delegation of decision-making authority to lower levels.

Solution

c

2. MC.11.02

Which of the following is *not* a reason for decentralizing?

a. Training and motivating managers

b. Unmasking inefficiencies in subdivisions of an overall profitable company

c. Allowing top management to focus on strategic decision-making

✓ d. Allowing top management to make all key operating decisions throughout the company

e. All of these.

Feedback

Check My Work

Reasons for decentralization include access to local information, cognitive limitations, more timely responses, focusing of central management, training, and motivation.

Solution

d

3. MC.11.03.ALGO (Algorithmic)

A responsibility center in which a manager is responsible only for costs is a(n)

a. revenue center.

✓ b. cost center.

c. profit center.

d. operations center.

Feedback

Check My Work

There are four responsibility centers: cost center (manager responsible only for costs), revenue center (manager responsible only for revenue), profit center (manager responsible for both revenues and costs), and investment center (manager responsible for revenues, costs, and investments).

Solution

b

4. MC.11.04.ALGO (Algorithmic)

Clariday Company has been tracking data on sales, costs, and investment for several years. They are analyzing the data to determine whether there are any trends in sales and costs that may be expected to continue into the next year. This is an example of using data analysis in which of the following ways?

✓ a. Predictively

b. Prescriptively

c. Diagnostically

d. Descriptively

Feedback

Check My Work

Correct

Solution

a

5. MC.11.05

If sales and average operating assets for Year 2 are identical to their values in Year 1, yet operating income is higher, Year 2 return on investment (compared with Year 1 ROI) will

a. decrease.

✓ b. increase.

c. stay the same.

d. The direction of change in ROI cannot be determined by this information.

Feedback

Check My Work

Margin = Operating Incom ÷ Sales

Turnover = Sales ÷ Average Operating Assets

ROI = Margin × Turnover

Solution

b

6. MC.11.06

If sales and average operating assets for Year 2 are identical to their values in Year 1, yet operating income is higher, Year 2 turnover (compared with Year 1 turnover) will

a. decrease.

b. increase.

✓ c. stay the same.

d. The direction of change in turnover cannot be determined by this information.

Feedback

Check My Work

Margin = Operating Incom ÷ Sales

Turnover = Sales ÷ Average Operating Assets

ROI = Margin × Turnover

Solution

c

7. MC.11.07.ALGO (Algorithmic)

The key difference between residual income and EVA is that EVA

a. must be greater than residual income.

b. There is no difference between residual income and EVA.

c. uses the ROI for a company rather than the actual percentage cost of capital.

d. is a ratio rather than an absolute dollar amount.

✓ e. uses the actual cost of capital for the company rather than a minimum required cost of capital.

Feedback

Check My Work

Residual Income = Operating Income − (Minimum Rate of Return × Average Operating Assets)
EVA (economic value added) requires the actual cost of capital to be used as minimum rate of return in residual income calculation. EVA always uses after-tax income.

Solution
e

8. MC.11.08.ALGO (Algorithmic)

If ROI for a division is 15% and the company's minimum required cost of capital is 18%, then

a. EVA must be negative.

✓ b. residual income for the division is negative.

c. residual income for the division equals the cost of capital.

d. EVA is greater than residual income.

e. residual income is positive.

Feedback

Check My Work

Residual Income = Operating Income − (Minimum Rate of Return × Average Operating Assets)
ROI = Operating Income/Average Operating Assets

Solution
b

9. MC.11.09

Division A manufactures an aircraft engine component with unit variable product cost of $38 and market price of $50. Division A incurs shipping costs of $3 per unit for sales to outside parties only. Division B uses this component in the manufacture of its own engine production activities. Top management allows negotiated transfer pricing. If Division A is operating at full capacity, the maximum transfer price (the ceiling of the bargaining range) is

a. $38.

✓ b. $50.

c. $44.

d. $47.

e. There is no bargaining range.

Feedback

Check My Work

The selling division sets the minimum price and the buying division sets the maximum price.

Solution
b

10. MC.11.10

Division A manufactures an aircraft engine component with unit variable product cost of $38 and market price of $50. Division A incurs shipping costs of $3 per unit for sales to outside parties only. Division B uses this component in the manufacture of its own engine production activities. Top management allows negotiated transfer pricing. If Division A is operating at less than full capacity, the minimum transfer price (the floor of the bargaining range) is

✓ a. $38.

b. $50.

c. $44.

d. $47.

e. There is no bargaining range.

Feedback

Check My Work

The selling division sets the minimum price and the buying division sets the maximum price.

Solution
a

11. MC.11.11

(Appendix 11A) Which of the following is a perspective of the Balanced Scorecard?

- a. Learning and growth (infrastructure)
- b. Internal business process
- c. Customer
- d. Financial
- ✓ e. All of these

Feedback

Check My Work

The four perspectives of the Balanced Scorecard are financial, customer, internal (business processes), and learning and growth.

Solution
e

12. MC.11.12

(Appendix 11A) The length of time it takes to produce a unit of output from the time raw materials are received until the good is delivered to finished goods inventory is called

- a. velocity.
- ✓ b. cycle time.
- c. manufacturing cycle efficiency.
- d. theoretical cycle time.
- e. theoretical MCE.

Feedback

Check My Work

The time to respond to a customer order is referred to as *responsiveness. Cycle time* and *velocity* are two operational measures of responsiveness.

Solution
b

13. MC.11.42

Which of the following is true of revenue centers?

- a. They are responsibility centers wherein managers are responsible for costs and quantity sold.
- ✓ b. They are responsibility centers wherein managers are responsible for price and quantity sold.
- c. They are responsibility centers wherein managers are responsible for investments.
- d. They are responsibility centers wherein managers are responsible for income.

Feedback

Check My Work
Correct
Post-Submission

In a revenue center, a manager is responsible only for sales, or revenue.

14. MC.11.43

A responsibility center in which a manager is responsible for both revenues and costs is a(n):

✓ a. profit center.

b. investment center.

c. revenue center.

d. cost center.

Feedback

Check My Work
Correct
Post-Submission

In a profit center, a manager is responsible for both revenues and costs.

15. MC.11.44

A responsibility center in which a manager is responsible for revenues, costs, and investments is a(n):

a. cost center.

b. revenue center.

✓ c. investment center.

d. profit center.

Feedback

Check My Work
Correct
Post-Submission

In an investment center, a manager is responsible for revenues, costs, and investments.

16. MC.11.45

The decision-making approach that allows managers at lower levels to make and implement key decisions pertaining to their areas of responsibility is:

✓ a. decentralization.

b. responsibility accounting.

c. optimal strategic accounting.

d. controllable accounting.

Feedback

Check My Work
Correct
Post-Submission

Decentralized decision making allows managers at lower levels to make and implement key decisions pertaining to their areas of responsibility.

17. MC.11.47

A segment of ABC, Inc. manufactures and sells blankets. The various models of blankets are produced in a single factory using stable technology. They are sold by the sales department, also located in the factory. The segment is most probably accounted for as a(n):

✓ a. profit center.

b. investment center.

c. revenue center.

d. cost center.

Feedback

Check My Work

Correct

Post-Submission

A profit center is the part of an organization in which the manager is responsible for both sales and costs.

18. MC.11.48

JetSky Airways has three divisions, the Western Division, the Eastern Division, and the Northern Division. The manager of the Western Division had wanted to purchase replacement airplanes for the division. However, he decided against it because although revenues would increase and the new planes would be less expensive to operate, the initial cost of the planes was quite large. The Western Division is most probably accounted for as a(n):

a. profit center.

b. cost center.

✓ c. investment center.

d. revenue center.

Feedback

Check My Work

Correct

Post-Submission

An investment center manager is responsible for revenues, costs, and investments.

19. MC.11.54

Division M had an ROI of 20% last year. The manager of Division M is considering an additional investment for the coming year. Which of these steps is the manager likely to take?

a. Reject the investment if it returns more than 20% ROI.

b. Accept the investment as long as it provides positive operating income.

c. Accept the investment as long as its ROI is positive.

✓ d. Reject the investment if it returns less than 20% ROI.

Feedback

Check My Work

Correct

Post-Submission

The hurdle rate (20%) indicates the minimum ROI necessary to accept an investment. If an investment's hurdle rate is less than 20%, management would reject the investment.

20. MC.11.58

Clothier Company had sales revenue and operating expenses of $5,000,000 and $4,300,000, respectively, for the year just ended. If average operating assets amounted to $6,000,000, the firm's ROI **(rounded to two decimal places)** was:

a. 13.33%.

b. 10.00%.

✓ c. 11.67%.

d. 83.33%.

Feedback

Check My Work

Correct

Post-Submission

Return on Investment = Operating Income ÷ Average Operating Assets = ($5,000,000 − $4,300,000) ÷ $6,000,000 = 11.67%

21. MC.11.59

Madison Company had an operating income of $14,000, average operating assets of $125,000, and sales of $50,000. What is Madison's return on investment (ROI)? **(Round answer to two decimal places.)**

a. 5.80%

b. 15.80%

c. 9.60%

✓ d. 11.20%

Feedback

Check My Work
Correct
Post-Submission

Return on Investment (ROI) = Operating Income ÷ Average Operating Assets = $14,000 ÷ $125,000 = 0.1120, or 11.20%

22. MC.11.60

Howard Company had an operating income of $75,000, sales of $200,000, and a turnover ratio of 0.52. What is Howard's return on investment (ROI)? **(Round answer to two decimal places.)**

a. 18.75%

b. 50.60%

✓ c. 19.50%

d. 32.50%

Feedback

Check My Work
Correct
Post-Submission

ROI = Margin × Turnover = (Operating Income ÷ Sales) × Turnover = ($75,000 ÷ $200,000) × 0.52 = 0.1950, or 19.50%

23. MC.11.61

Omega Division's data are as follows:

Average operating asset base	$500,000
Operating income	$75,000
Cost of capital	14%
Target return on investment (ROI)	16%
Margin	20%

If the asset base is decreased by $120,000, with no other changes, what will Omega Division's return on investment be? **(Round answer to two decimal places.)**

a. 10.50%

✓ b. 19.74%

c. 18.50%

d. 15.79%

Feedback

Check My Work
Correct
Post-Submission

SUPPORTING CALCULATIONS:

Alpha's ROI = Operating Income ÷ (Average Operating Asset Base − Decrease in Asset Base) = $75,000 ÷ ($500,000 − $120,000) = 19.74%

24. MC.11.63

If the margin of 0.30 stayed the same and the turnover of 5.0 increased by 12%, the ROI would:

a. decrease by 12%.

b. increase by 15%.

✓ c. increase by 12%.

d. remain the same.

Feedback

Check My Work
Correct
Post-Submission

SUPPORTING CALCULATIONS:
[(0.30 × 5.6*) − 1.5**] ÷ 1.5 = 12% increase

*5.0 × 1.12
**0.30 × 5.0

25. MC.11.68

Align Company provided the following information for last year:

Operating income	$90,000
Sales	240,000
Beginning operating assets	410,000
Ending operating assets	440,000

Calculate Align's margin for last year. **(Round answer to two decimal places.)**

a. 0.40

b. 0.50

c. 0.26

✓ d. 0.38

Feedback

Check My Work
Correct
Post-Submission

Margin = Operating Income ÷ Sales = $90,000 ÷ $240,000 = 0.38

26. MC.11.71

The following information pertains to the three divisions of Yancy Company:

	Division A	Division B	Division C
Sales	?	?	$1,345,000
Net operating income	$48,000	$18,000	$85,000
Average operating assets	$420,000	?	?
Return on investment	?	15%	20%
Margin	0.200	0.015	?
Turnover	2.1	?	?
Target ROI	17%	14%	8%

What are the average operating assets for Division C?

a. $95,000

b. $420,000

c. $82,000

✓ d. $425,000

Feedback

Check My Work
Correct
Post-Submission

$85,000 ÷ 0.20 = $425,000

27. MC.11.72

The following information pertains to the three divisions of Yancy Company:

	Division A	**Division B**	**Division C**
Sales	?	?	$1,345,000
Net operating income	$48,000	$18,000	$85,000
Average operating assets	$420,000	?	?
Return on investment	?	15%	20%
Margin	0.200	0.015	?
Turnover	2.1	?	?
Target ROI	17%	14%	8%

What is the turnover for Division C? **(Round answer to one decimal place.)**

✓ a. 3.2

b. 1.5

c. 0.2

d. 6.7

Feedback

Check My Work
Correct
Post-Submission

Average Operating Assets = $85,000 ÷ 0.20 = $425,000
Turnover = $1,345,000 ÷ $425,000 = 3.2 times

28. MC.11.82

Which of the following is a percentage, not an absolute dollar measure?

a. Residual income

✓ b. Return on investment

c. Operating income

d. Economic value added (EVA)

Feedback

Check My Work
Correct
Post-Submission

All the given measures, except return on investment, are expressed in terms of dollars and not in terms of percentages.

29. MC.11.83

Economic value added is residual income with the cost of capital equal to the firm's:

a. standard cost of capital.

b. budgeted cost of capital.

✓ c. actual cost of capital.

d. average cost of capital.

Feedback

Check My Work

Correct

Post-Submission

Basically, EVA is residual income with the cost of capital equal to the actual cost of capital for the firm (as opposed to some minimum rate of return desired by the company for other reasons).

30. MC.11.86

Using economic value added (EVA) to calculate residual income, the cost of capital employed is the:

a. actual percentage cost of capital multiplied by the average capital employed.

b. standard percentage cost of capital multiplied by the total capital employed.

✓ c. actual percentage cost of capital multiplied by the total capital employed.

d. standard percentage cost of capital multiplied by the average capital employed.

Feedback

Check My Work

Correct

Post-Submission

EVA is after-tax operating income minus the dollar cost of capital employed. The dollar cost of capital employed is the actual percentage cost of capital multiplied by the total capital employed.

31. MC.11.87

The following information pertains to the three divisions of Morning Company:

	Division X	**Division Y**	**Division Z**
Sales	?	?	$1,250,000
Net operating income	$40,000	$25,000	$75,000
Average operating assets	$300,000	?	?
Return on investment	?	20%	15%
Margin	0.10	0.05	?
Turnover	1.2	?	?
Target ROI	15%	12%	10%

What is the residual income for Division X?

a. $9,000

✓ b. $(5,000)

c. $5,000

d. $(9,000)

Feedback

Check My Work

Correct

Post-Submission

SUPPORTING CALCULATIONS:

$40,000 − ($300,000 × 0.15) = $(5,000)

32. MC.11.95

Several transfer pricing policies are used in practice that include:

a. transfer at negotiated price.

b. transfer at cost.

c. transfer at market price.

✓ d. All of these are correct.

Feedback

Check My Work

Correct

Post-Submission

The transfer pricing policies used in practice are market price, cost-based transfer price, and negotiated transfer price.

33. MC.11.96

Division A produces a component and wants to sell it to Division B. The transfer price is a:

a. cost to Division "B" and has no effect on Division A.

b. revenue to Division "A" and has no effect on Division B.

c. revenue to Division "B" and a cost to Division A.

✓ d. revenue to Division "A" and a cost to Division B.

Feedback

Check My Work

Correct

Post-Submission

The value of the transferred good is a revenue to the selling division and a cost to the buying division.

34. MC.11.99

Prestigious, Inc. has many divisions that are evaluated on the basis of return on investment (ROI). One division, Kappa, makes boxes. A second division, Alpha, makes chocolates and needs 90,000 boxes per year. Kappa incurs the following costs for one box:

Direct materials	$0.40
Direct labor	0.70
Variable overhead	0.50
Fixed overhead	0.15
Total	$1.75

Kappa has the capacity to make 720,000 boxes per year. Alpha currently buys its boxes from an outside supplier for $2.00 each (the same price that Kappa receives). Assume that Prestigious, Inc. mandates that any transfers take place at full manufacturing cost. What would be the transfer price if Kappa transferred boxes to Alpha?

a. $1.00

✓ b. $1.75

c. $1.35

d. Cannot be determined from the given information

Feedback

Check My Work

Correct

Post-Submission

Full Cost for Producing a Box = Direct Materials + Direct Labor + Variable Overhead + Fixed Overhead = $0.40 + $0.70 + $0.50 + $0.14 = $1.75

35. MC.11.102

Prestigious, Inc. has many divisions that are evaluated on the basis of return on investment (ROI). One division, Kappa, makes boxes. A second division, Alpha, makes chocolates and needs 90,000 boxes per year. Kappa incurs the following costs for one box:

Direct materials	$0.40
Direct labor	0.70
Variable overhead	0.50
Fixed overhead	0.15
Total	$1.75

Kappa has the capacity to make 720,000 boxes per year. Alpha currently buys its boxes from an outside supplier for $2.00 each (the same price that Kappa sells its product for). Assume that Prestigious, Inc. allows division managers to negotiate the transfer price. Kappa is currently producing only 650,000 boxes. If Kappa and Alpha agree to transfer boxes, what is the ceiling of the bargaining range, and which division sets it?

- a. $1.35; Kappa
- ✓ b. $2.00; Alpha
- c. $1.35; Alpha
- d. $1.48; Kappa

Feedback

Check My Work

Correct

Post-Submission

As the buyer, Alpha sets the ceiling of the bargaining range. The ceiling would be set at the market price of $2.00.

36. MC.11.105

Letters, Inc. has a number of divisions. One division, Beta, makes zippers that are used in the manufacture of boots. Another division, Lambda, makes boots that use the zippers and needs 95,000 zippers per year. Gamma incurs the following costs for one zipper:

Direct materials	$0.25
Direct labor	0.22
Variable overhead	0.98
Fixed overhead	1.40
Total	$2.85

Beta has the capacity to make 960,000 zippers per year, but because of a soft market, it only plans to produce and sell 630,000 zippers next year. Lambda currently buys zippers from an outside supplier for $4.00 each (the same price that Beta sells its product for).

Assume that Letters allows negotiated transfer pricing. What is the floor of the bargaining range, and which division sets it?

- a. $1.75; Lambda
- b. $2.70; Lambda
- ✓ c. $1.45; Beta
- d. $2.70; Beta

Feedback

Check My Work

Correct

Post-Submission

Being the seller, Beta sets the floor of the bargaining range. As Beta has excess capacity, it must cover its incremental costs, which are equal to variable manufacturing costs.

Minimum Transfer Price = Direct Materials + Direct Labor + Variable Overhead = $0.25 + $0.22 + $0.98 = $1.45

37. MC.11.111

The Balanced Scorecard perspective that describes the internal processes needed to provide value for customers and owners is the ______ perspective.

a. learning and growth

b. financial

✓ c. internal business process

d. customer

Feedback

Check My Work

Correct

Post-Submission

The internal business process perspective describes the internal processes needed to provide value for customers and owners.

38. MC.11.114

The Balanced Scorecard perspective that defines the capabilities that an organization needs to create long-term growth and improvement is the ______ perspective.

a. internal business process

b. customer

✓ c. learning and growth

d. financial

Feedback

Check My Work

Correct

Post-Submission

The learning and growth (infrastructure) perspective defines the capabilities that an organization needs to create long-term growth and improvement.

39. MC.11.115

A testable strategy is defined as a:

a. means of specifying objectives, measures, targets, and initiatives for each perspective of the Balanced Scorecard.

b. means of providing managers with information about the effectiveness of strategy implementation and the validity of th
underlying the strategy.

✓ c. set of linked objectives aimed at an overall goal.

d. strategic management system that defines a strategic-based responsibility accounting system.

Feedback

Check My Work

Correct

Post-Submission

A testable strategy can be defined as a set of linked objectives aimed at an overall goal.

40. MC.11.117

The process value chain consists of:

a. innovation, operations, and post-sales service.

b. innovation, satisfaction, and value.

c. design, production, and selling.

d. value, operations, and objectives.

Feedback

Check My Work
Correct
Post-Submission

The process value chain is made up of three processes: innovation process, operations process, post-sales service process.

41. MC.11.120

Last week, Davis worked for 50 hours and produced 250 business card boxes. What was the cycle time for producing one business card box?

a. 90 minutes

b. 4 minutes

c. 10 minutes

d. 12 minutes

Feedback

Check My Work
Correct
Post-Submission

Cycle Time = (Productive Hours in 1 Week × 60 Minutes per Hour) ÷ Actual Units Produced in a Week = (50 × 60) ÷ 250 = 12 minutes

42. MC.11.121

Last week, James worked for 40 hours and produced 160 business card boxes. What is the velocity of business card boxes per hour?

a. 4 units per hour

b. 6 units per hour

c. 3 units per hour

d. 10 units per hour

Feedback

Check My Work
Correct
Post-Submission

Actual Velocity = Actual Units Produced in a Week ÷ Productive Hours in 1 Week = 160 ÷ 40 = 4 units per hour

43. MC.11.125

Two Girls Sweets earned operating income last year as shown in the following income statement:

Sales	$480,000
Cost of goods sold	222,000
Gross margin	258,000
Selling and administrative expense	210,000
Operating income	$48,000

At the beginning of the year, the value of operating assets was $275,000. At the end of the year, the value of operating assets was $320,000. Two Girls Sweets requires a minimum rate of return of 10%. Calculate residual income.

a. $20,000

b. $10,000

✓ c. $18,250

d. $25,000

Feedback

Check My Work
Correct
Post-Submission

Average Operating Assets = (Beginning Assets + Ending Assets)/2 = ($275,000 + $320,000)/2 = $297,500
Residual Income = Operating Income − (Minimum Rate of Return × Average Operating Assets) = $48,000 − (0.10 × $297,500) = $18,250

44. MC.11.129

A Dance Studio earned operating income last year as shown in the following income statement:

Sales	$480,000
Cost of goods sold	220,000
Gross margin	250,000
Selling and administrative expense	210,000
Operating income	$50,000
Less: Income taxes (at 30%)	15,000
Operating income	$35,000

Total capital employed equals $200,000. A Dance Studio's actual cost of capital is 12%. Calculate EVA for A Dance Studio.

a. $10,000

✓ b. $11,000

c. $9,600

d. $8,000

Feedback

Check My Work
Correct
Post-Submission

EVA = After-Tax Operating Income − (Actual Percentage Cost of Capital × Total Capital Employed) = $35,000 × (0.12 x $200,000) = $35,000 − $24,000 = $11,000

45. MC.11.136

Which of the following is an example of the financial perspective of a Balanced Scorecard?

a. Percentage of employees participating in training courses

b. Employee turnover rate

c. Number of monthly complaints

✓ d. Total daily sales

Feedback

Check My Work
Correct
Post-Submission

Total daily sales is an example of the financial perspective of a Balanced Scorecard.

46. MC.11.137

Which of the following is an example of the customer perspective of a Balanced Scorecard?

a. Total daily sales

b. Percentage of employees participating in training courses

c. Employee turnover rate

✓ d. Number of monthly complaints

Feedback

Check My Work

Correct

Post-Submission

Number of monthly complaints is an example of the customer perspective of a Balance Score Card.

47. MC.11.138

Which of the following is an example of the internal perspective of a Balanced Scorecard?

a. Total daily sales

b. Percentage of employees participating in training courses

c. Number of monthly complaints

✓ d. Employee turnover rate

Feedback

Check My Work

Correct

Post-Submission

Number of monthly complaints is an example of the customer perspective of a Balance Score Card.

48. MC.11.139

Which of the following is an example of the learning and growth perspective of a Balanced Scorecard?

a. Number of monthly complaints

✓ b. Percentage of employees participating in training courses

c. Total daily sales

d. Employee turnover rate

Feedback

Check My Work

Correct

Post-Submission

The percentage of employees participating in training courses is an example of the learning and growth perspective of a Balanced Scorecard.